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What a deep song: The role of music features in perceived depth

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Abstract

This study examines perceptions of music depth by exploring its relationships to different music features. First, a correlation analysis shows that the perceived depth of music is negatively correlated with valence and arousal and is also related to different music features, including tempo, Mel-frequency cepstrum coefficients, chromagrams, spectral centroids, spectral bandwidth, spectral contrast, spectral flatness, spectral roll-off, and tonal centroid features. Applying machine learning methods, we find that selected music features can predict perceptions of music depth, and a random forest regression (RFR) is found to perform best in this study. Finally, a feature importance analysis shows that the principal component of spectral contrast dominates the RFR-based music depth recognition model, showing that deep music usually has clear and narrow-band audio signals.

KEYWORDS

computational modeling, machine learning, music depth recognition, music features, perceived depth

INTRODUCTION

From its ability to convey feelings, music plays a significant role in human life. To determine how music resonates with listeners, composers and researchers have explored the links between music features and music perceptions for decades (Feng et al., 2003; Yang et al., 2018). Various perceived psychological attributes of music such as valence (Dean & Bailes, 2016), complexity (Conley, 1981), and arousal (Gabrielsson & Lindström, 2001) have been explored by analyzing original music features. Such works serve as strong references for understanding of the internal mechanisms of music perception attributes, which may affect music preferences (Arkes et al., 1986), absorption (Herbert, 2013), music therapy (Dingle et al., 2015), and so forth.

From perceptions of 38 psychological attributes identified from different music excerpts, Greenberg et al. (2016) proposed three principal components (arousal, valence, and depth) of music perception. These principal components reflect the emotional (happy, sad, and angry) and cognitive (intelligent and sophisticated) aspects of the music which explicitly reflect individuals' personality features to some extent. Although arousal and valence have been widely explored in previous studies (Juslin & Laukka, 2004; Schuller et al., 2010), depth is rarely discussed. Greenberg et al. (2016) found intelligent, sophisticated, inspiring, complex, poetic, deep, emotional, and thoughtful attributes to be heavily loaded on the "depth" component while party music and danceable attributes are negatively loaded. Notably, although there are many other description models of music emotion (e.g., Zentner et al., 2008), the model of Greenberg et al. (2016) is mainly suitable for describing perceived emotions of music. In fact, only a few sub-attributes of depth have been studied. For instance, Nagathil et al. (2018) constructed a music feature-based linear regression model to predict music complexity, which was defined as how difficult the music was to follow and understand; Doi et al. (2018) investigated the effects of sophisticated music on males; and Rigg (1964) found that dreamy music tends to be slow and soft and to contain little dissonance. However, music depth has not been systematically investigated, and it still lacks a clear definition. Based on the result of Greenberg et al. (2016), the present study regarded music depth as a psychological perceptual attribute for evaluating how deep music is; while deep music represents a type of music, which is the opposite of dance music and party music, and is perceived as thoughtful, complex, intelligent, sophisticated, inspiring, poetic, or emotional. In reference to research on sub-attributes of depth, the present study explores the relations between perceptions of music depth and different music features.

Music features can be divided into three levels. First, low-level music features such as Mel-frequency cepstrum coefficients (MFCC; Wang et al., 2012), chromagram (Schmidt et al., 2010), and the low energy rate (Lu et al., 2005) are extracted from raw music data, and they are directly related to the human auditory system (Yang et al., 2018). These features have been widely applied for music signal processing tasks, including those involving music emotion recognition (Lu et al., 2005), music genre classification (Meng et al., 2007), optical music recognition (Rebelo et al., 2012), and so forth. Next, low-level music features are commonly used to represent mid-level features (Xu et al., 2020a) such as timbre, intensity, and tempo. Since mid-level music features are well known in conventional musical writing, they are more widely considered in music psychology research (e.g., Gomez & Danuser, 2007; Juslin & Laukka, 2004). Third, high-level music features related to the human cognitive system have attracted substantial attention from both music psychologists and computer scientists. Different high-level music features such as perceived emotion (Yang et al., 2018), felt emotion (Schubert, 2013), and depth have motivated unique research focuses. Low- and mid-level music features are usually applied to describe high-level features (e.g., Vempala & Russo, 2018). In this study, the perceived depth of music is analyzed using a music feature-based method, and all three levels are considered.

Many studies have investigated the linear relationships between high-level music features and different music dimensions and examined the consistency of these relations (e.g., Gabrielsson & Juslin, 2003; Nagathil et al., 2018; Rigg, 1964). However, the relationships between variables may not only be linear (Xu et al., 2020a, 2021b). Applying machine learning (ML) approaches, Vempala and Russo (2018) conducted a relationship investigation from a computational modeling perspective, leading to new lines of music cognition research. With this approach, various association forms can be studied using different ML methods, including the k-nearest neighbor (Dewi & Harjoko, 2010), support vector regression (SVR; Deng et al., 2015), and random forest (Song et al., 2012) methods. This method is also of practical value because constructed ML models can automatically recognize music perception information. The above information has also been used for music recommendation (Park et al., 2015; Xu et al., 2021a) and music retrieval (Downie, 2008). Thus, the present study considers both linear statistics and ML methods for a more comprehensive analysis.

In summary, music psychologists have made great progress on the connection between musical structure (music features) and the perceived emotion of music. Individuals are drawn to musical styles with particular social connotations, such as toughness, rebellion, distinctiveness, and sophistication (Abrams, 2009; Schwartz & Fouts, 2003; Tekman & Hortacsu, 2002), indicating that the psychological responses that music brings to an individual included not only emotional responses but also cognitive processes (including perceived depth) (Greenberg et al., 2016). However, the association between music features and perceived depth of music has rarely been studied. Therefore, the main purpose of this research is to investigate the association between music structure (music features) and music depth. Referring to previous music emotion studies, we adopted three steps to achieve our goal. First, we investigated the relationship between high-level music features (arousal, valence, and depth). According to the findings of Greenberg et al. (2016), some sub-attributes of depth loaded highly on the "depth" component (e.g., poetic and dreamy) are negatively loaded on the "valence" and "arousal" components. Thus, we assume that perceived valence and arousal are negatively correlated with perceived depth of music (Hypothesis 1). Second, we investigated the relationship between music depth and low-level or mid-level music features. As previous works have shown that some sub-attributes of depth are correlated with different low- and mid-level music features (e.g., tempo; Rigg, 1964), we assume that the perception of music depth is also correlated with different music features (Hypothesis 2). The considered low- and mid-level music features are introduced in more detail in Section 2.2. Finally, to investigate the nonlinear relationships between variables, we construct music depth recognition models by using music features as inputs to predict perceived depth values (ground truth data). Previous studies have shown that nonlinear ML algorithms perform well for other music perception recognition tasks (e.g., Dong et al., 2019). Thus, we assume that our depth recognition models based on nonlinear ML methods should perform better than those using the linear method (Hypothesis 3), denoting that nonlinear relationship can better explain the links between music features and perceived depth.

METHODS

This study was conducted based on the PSIC3839 data set, a public-free data set for music emotion recognition studies (Xu et al., 2020b). This data set contains manually annotated arousal, valence, and depth scores for 3,839 popular songs in China as well as the original music files. Data processing and analysis methods used are introduced in this section. Section 2.1 introduces the PSIC3839 data set by describing its basic information and data distribution. Section 2.2 introduces the low- and mid-level music features extracted from the music files. Then, Section 2.3 introduces how the recognition models are constructed.

PSIC3839 data set

The PSIC3839 data set was compiled through an online experiment (Xu et al., 2020b). Each song was annotated by five to seven unique participants, and mean scores for arousal, valence, and depth were published. The basic statistics for the PSIC3839 data set are shown in Table 1. The distribution of the depth dimension shows a right-skewed distribution, and the valence dimension presents as a fuzzy bipolar distribution. The arousal dimension shows a peak at [–0.5,0], dispersing from the middle to both sides. A total of 85 annotators (61.2%

TABLE 1 The basic statistics for PSIC3839 data set ($n = 3839$)

	Mean	*SE*	Median	Skewness	Kurtosis
Arousal	0.197	0.016	0.167	–0.034	–1.014
Valence	0.045	0.017	0.000	0.049	–1.299
Depth	0.259	0.016	0.500	–0.561	–0.758

females) aged 21.92 years ($SD = 2.03$) were included in the study. Of the annotators, 96.47% reported that they often listened to music and eight participants had professional musically trained.

Before annotating, participants were introduced the meanings of the words ("valence", "arousal", and "depth") by describing their corresponding adjectives. The correspondence between the different dimensions and adjectives used were according to the findings in Greenberg et al. (2016). Participates were informed that high-depth music can be described as intelligent, sophisticated, inspiring, complex, poetic, dreamy, thoughtful, and emotional; and low-depth music can be described as party music and danceable. The evaluation of arousal and depth was conducted on a 5-point Likert scale from –2 (*not at all*) to 2 (*very much*), and valence was measured at –2 (*negative*) to 2 (*positive*). To ensure the quality of the annotation, participants were asked to conducted listening tasks with eyes closed and in a quiet place. In addition, the *SD* of each song is published as an indicator of scoring consistency. All the procedures involved in this study are in line with the ethical standards of Academic Committee of Zhejiang University and the 1964 Helsinki Declaration.

Music feature extraction

To explore the relations between music features and perceived depth, we selected music features based on previous work in music psychology (Gabrielsson & Juslin, 2003; Gomez & Danuser, 2007) and music signal processing (McFee et al., 2018; Yang et al., 2018). Using the Librosa package (McFee et al., 2015) for feature extraction, the following nine types of features were considered:

1. Timbre-related features: four features, including the spectral centroid, bandwidth, roll-off, and flatness, are extracted from the power spectrogram of music and are usually related to tone structure (Soleymani et al., 2013).
2. Pitch-related features: chromagrams can describe the pitch components of music over a short time interval (Schmidt et al., 2010) and are determined from a power spectrogram using Ellis' (2007) method.
3. Harmonic-related features: tonal centroid features (tonnetz) are useful for harmonic change detection (Harte et al., 2006) and are calculated by projecting chromagram features onto a 6-dimensional basis.
4. Tempo: we estimate tempo (beats per minute) by analyzing original audio.
5. Mel-frequency cepstrum coefficients reflect the nonlinear frequency sensitivity of the human auditory system and can be extracted from a mel-scaled spectrogram (Wang et al., 2012).
6. Spectral contrast: this is an effective feature for music genre classification (Jiang et al., 2002). High contrast values generally correspond to clear, narrow-band signals while low contrast values correspond to broad-band noise (McFee et al., 2015).

Since most of the studied features were extracted from the time series, their mean values were first calculated for linear analysis. Each continuous feature was scaled to a value of 0 to 1 via min–max scaling (Kahng et al., 2002). Similarly, the original depth values were scaled to values of 0 to 1 for data presentation and for subsequent analysis.

Model construction

Further, we explore the effect of music features on perceived depth by building music depth recognition models. Figure 1 shows the model construction process. First, the power spectrogram of each song was extracted by computing

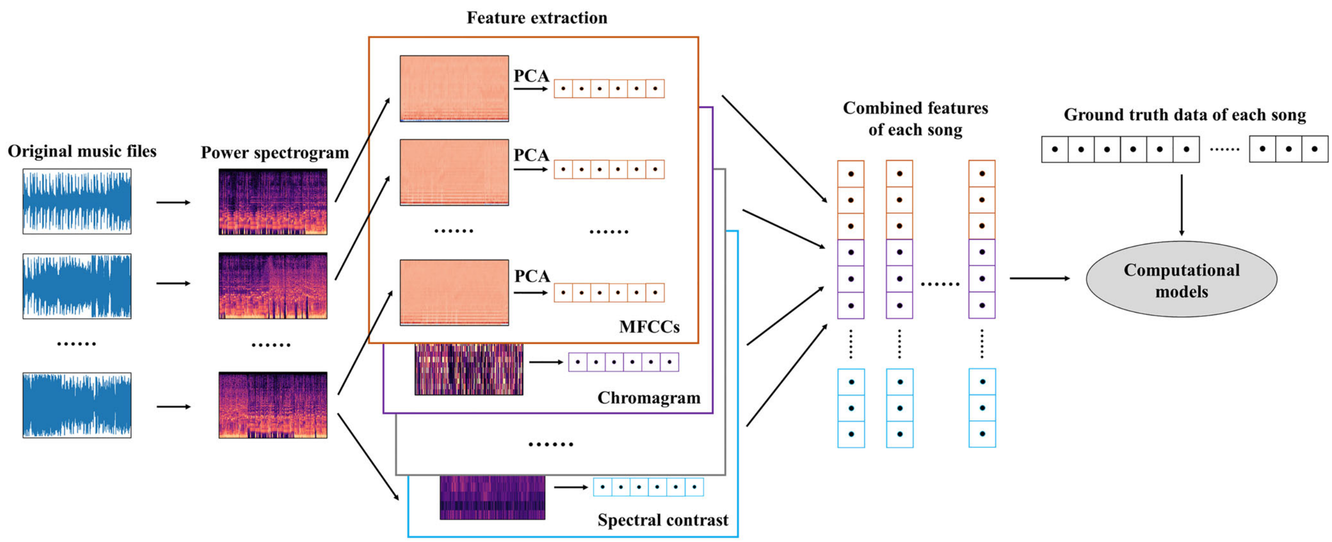

FIGURE 1 The model construction process

discrete Fourier transforms over short overlapping windows. Second, different music features (mentioned in Section 2.2) were extracted from the time series using the Librosa package (McFee et al., 2015). Third, we used a principal components analysis (PCA) rather than calculating mean values (Section 2.2) to reduce the dimensionality of features because a PCA can retain the timing information of features. In addition, to preserve the interpretability of features, we separately reduced the dimensions of each type of feature, though reducing the dimensions of all features together can better reduce computational and storage space (Vempala & Russo, 2018; Xu et al., 2020a). After conducting the PCA, 50 dimensions of each type of music feature (except for tempo) were retained, and these dimensions were combined as the final model inputs. Finally, using ML methods to link the inputs and ground truth data (perceived depth values), music depth recognition models were constructed.

We formulated music depth recognition as a regression problem (predicting a real value from observed features; Sen & Srivastava, 2012), and three traditional ML methods were considered. We first used multiple linear regression (MLR) as a baseline method given its low computational complexity and high interpretability (Franklin, 2005). SVR and random forest regression (RFR) were then applied to improve the model effects because they have been found in many cases to be superior to existing ML methods (e.g., Bai et al., 2016; Yang et al., 2007).

We built three separate models using different ML methods to predict perceptions of music depth. A grid parameter search was used to find the best parameters for each model

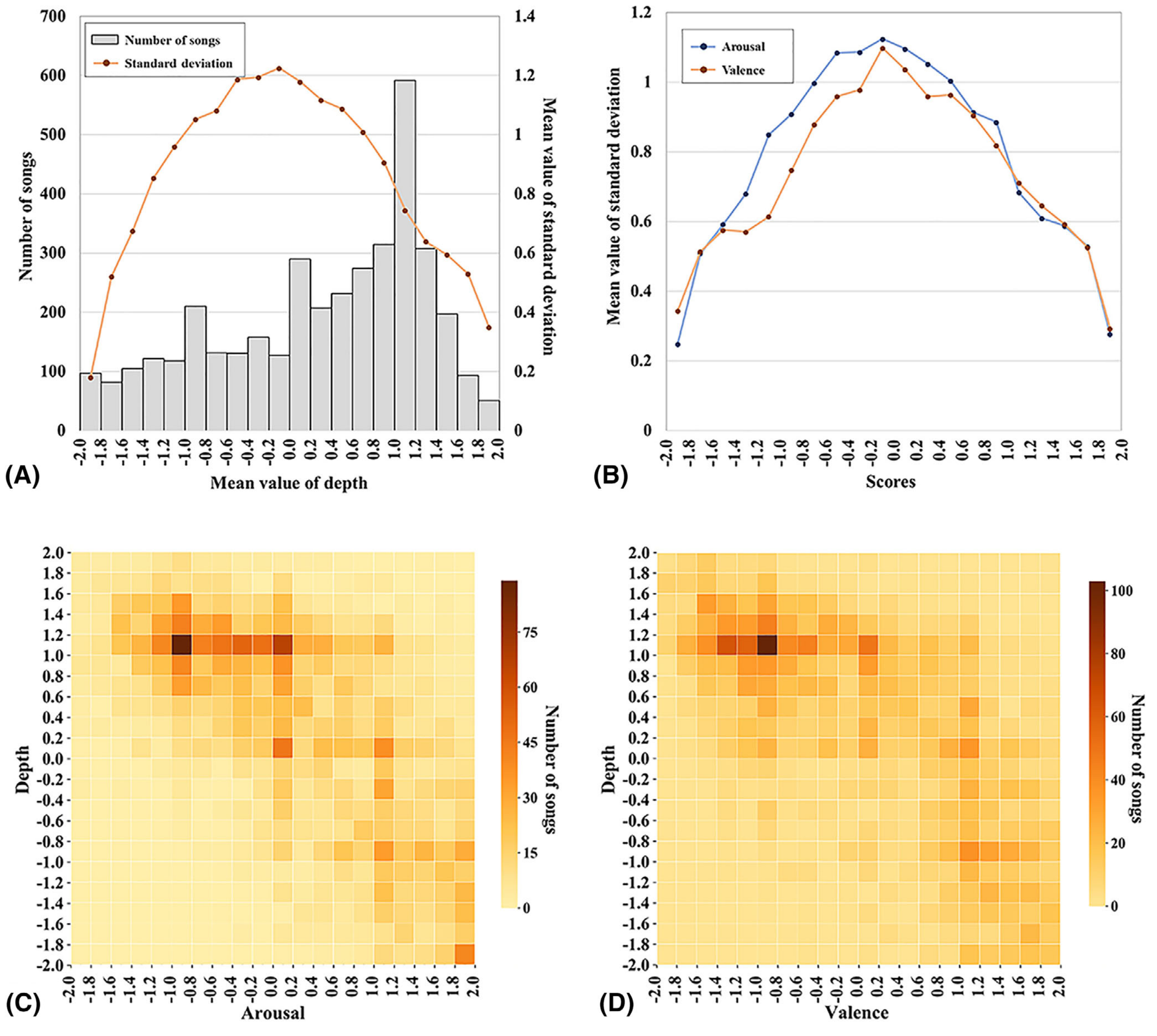

FIGURE 2 The distribution of scores for all songs in the PSIC3839 data set

(Yang et al., 2007). The performance of the models was evaluated by the 10-fold cross-validation technique, which uses 90% data as training data to train models and the remaining instances as testing data. Model prediction accuracy was measured in terms of the root-mean-square error (RMSE) and Pearson's correlation coefficient (PCC) (Dong et al., 2019; Gladstone et al., 2019).

RESULTS

Relationships between high-level music features

Figure 2A shows the distribution of depth scores, and we found that most songs were evaluated as deep music (mean = 0.26, *SD* = 0.99; 62.67% of music scores are >0). While, even with the same music, different people may experience different perceived emotions (Gabrielsson & Lindström, 2001). The *SD* represents the individuals' consistence of the perceived emotions. The mean *SD* of songs of different score intervals shows an inverted U-shaped distribution, and a similar *SD* distribution appears in the arousal and valence scores (see Figure 2B). We further investigate the relations between mean scores and *SD* values to explore this consistence difference. A correlation analysis shows that the depth scores are negatively correlated with *SD* values [r (3,838) = −.14, $p < .001$], meaning that people's evaluations may diverge as music depth decreases.

Figure 2C,D show the distribution of songs in the arousal-depth and valence-depth spaces, respectively. A large proportion of the songs fall under the fourth quadrant followed by the first and second quadrants. The perceived depth scores are negatively correlated with arousal [r (3,838) = −.630, $p < .001$] and valence [r (3,838) = −.628, $p < .001$]. As shown in Table 2, there are significant differences between the depth scores of the songs in the top 27% and bottom 27% groups in both arousal and valence according to independent sample t test [t (2,068) = 48.61, $p < .01$, Cohen's $d = 2.14$, $r = .73$; t (2,067.98) = 49.908, $p < .01$, Cohen's $d = 2.19$, $r = .74$]. Negative or low-arousal music is more likely to be perceived as deep music, supporting Hypothesis 1. However, this result may be explained by the relations between high- and mid-level music features. For example, dreamy music (a sub-attribute of depth) is more likely to be slow and soft (Rigg, 1964) while high arousal music is more likely to be fast and exuberant (Gabrielsson & Lindström, 2001), which may cause their negative correlation. Thus, in our next analysis, we examine music depth in relation to mid- and low-level features.

TABLE 2 Differences in arousal and valence between high depth songs and low depth songs

	High depth		Low depth	
	Mean	***SD***	**Mean**	***SD***
Arousal	−0.492	0.784	1.076	0.680
Valence	−0.729	0.780	0.980	0.778

Relationships between depth and mid- or low-level music features

Figure 3 shows the distribution of songs in different two-dimensional space. We found that in each space, most of the songs fall within a few squares; thus, the perceived depth values are related to the considered music features. A correlation analysis was then conducted (see Table 3). We found depth values to be positively correlated with spectral contrast [r (3,838) = .590, $p < .001$], meaning that deep music usually has clear and narrow-band audio signals. Mean chromagram values were found to be negatively correlated with perceived depth [r (3,838) = −.532, $p < .001$], denoting that deep music is more likely to be low pitched. Although Hypothesis 2 is supported, not all of the results can be explained by a linear relationship. Thus, ML methods were applied for subsequent analysis by building computational models.

The performance of ML models

Root-mean-square error and PCC values for the different models are shown in Figure 4. We found the baseline model using the MLR method to reach a mean PCC value of 0.700 and a mean RMSE value of .180. The SVR-based recognition model reached a mean PCC value of 0.714 and a mean RMSE value of .176, which are significantly better than those of the MLR-based model (PCC: $t = 3.282$, $p < .01$, $d = 0.584$; RMSE: $t = -2.746$, $p < .01$, $d = -0.203$). Finally, the RFR-based model performed best, achieving a mean PCC value of 0.722 and a mean RMSE value of .173. The above results support Hypothesis 3; that is, nonlinear relationships may better explain the links between music features and perceived depth. RFR, which involves creating an ensemble of decision trees to predict the perceived depth, was identified as the most suitable method.

We then attempted to interpret our models by examining the information gains of features. Models constructed by MLR can be interpreted by feature coefficients, and models constructed by RFR can be interpreted by calculating feature importance (Quiroz et al., 2018). However, SVR-based models cannot be interpreted in the above manner, so only MLR- and RFR-based models are considered here. In addition, since we use 10-fold cross-validation to evaluate model performance, the feature weight may differ when predicting different test sets; thus, we use boxplots to plot the distribution of feature weight (Figure 5). We find the first PCA component of spectral contrast and the fifth PCA component of spectral flatness to play the most important roles in the MLR-based model (Figure 5A), reaching mean coefficients of 0.675 and −0.669, respectively. From the absolute values of coefficients of the same types of features (e.g., the PCA components of MFCC), spectral

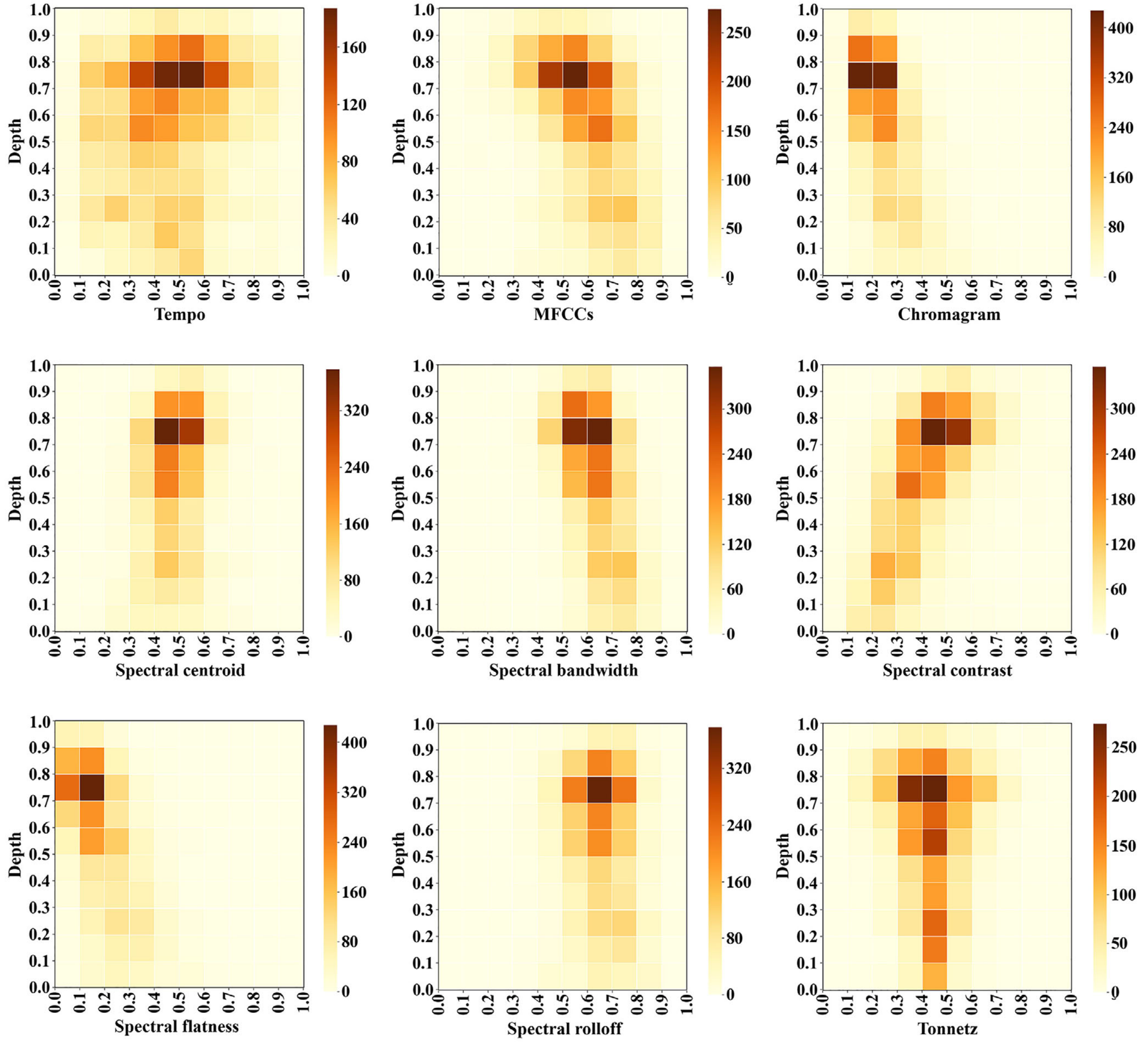

FIGURE 3 Distribution of songs in two-dimensional spaces

TABLE 3 Correlation between depth and music features.

	Tempo	MFCC	Chroma	Cent	BW	Contrast	Flatness	Roll-off	Tonnetz
Depth	0.128***	–0.433***	–0.532***	0.174***	–0.442***	0.590***	–0.553***	–0.099***	–0.057***

Abbreviations: BW = spectral bandwidth; Cent = spectral centroid; Chroma = chromagram; Contrast = spectral contrast; Flatness = spectral flatness; MFCC = Mel-frequency cepstrum coefficients; Roll-off = spectral roll-off; Tonnetz = tonal centroid features.

***Correlation is significant at the .001 level (2-tailed).

flatness accounts for 19.85% of the MLR-based model followed by spectral roll-off (15.47%), spectral centroid (14.47%), MFCC (13.32%), spectral bandwidth (12.28%), spectral contrast (10.73%), chromagram (6.97%), tonnetz (6.67%), and tempo features (0.26%). For the RFR-based model (Figure 5B), the first PCA component of spectral contrast is also important; it even occupies a dominant position, accounting for 70.04% of the model. Similarly, from the feature importance of the same types of features, we find that spectral contrast accounts for 74.96% of the RFR-based model followed by MFCC (7.51%), spectral bandwidth (5.52%), spectral flatness (3.86%), tonnetz (2.73%), chromagram (2.34%), spectral centroid (1.54%), spectral roll-off (1.33%), and tempo features (0.21%). The

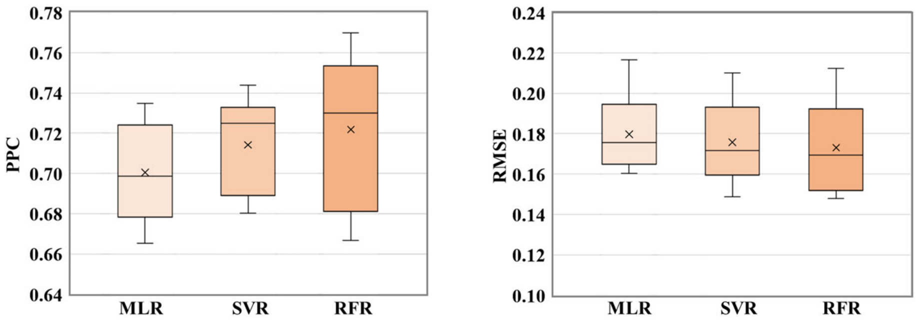

FIGURE 4 Performance of music depth recognition models adopting different machine learning (ML) methods. Each box shows the distribution of the predictive results of 10 test sets. The black line in the middle of each box indicates the median; "×" indicates the mean value

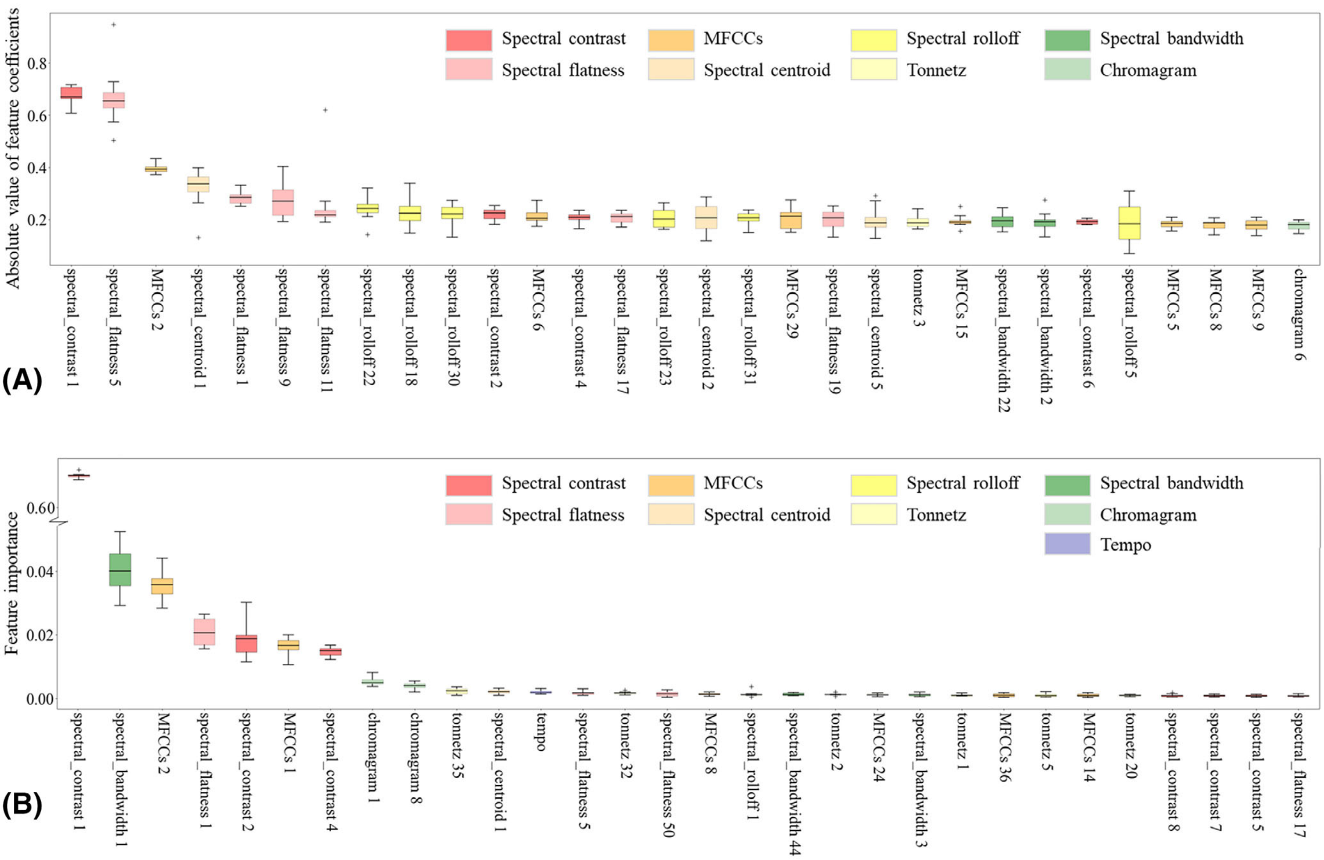

FIGURE 5 Distribution of importance of each feature for different recognition models. Boxplots are sorted by median values, and only the top 30 features are included for visibility with trends of the remaining features being approximately the same; the black line in the middle of each box indicates the median. (A) Absolute value of feature coefficients of the machine learning (ML)-based model. (B) Feature importance of the random forest regression (RFR)-based model

different proportions of feature importance reveal differences between the MLR and RFR algorithms in interpreting perceptions of music depth.

Finally, we explore the meaning of the decisive feature (the first PCA component of spectral contrast) in the RFR-based model. PCA shows that the first component of spectral contrast explains 10.26% of the variation. Since sub-dimensions determined before the PCA represent spectral contrast values of the time series, we calculate the contributions of the values to the first component at different time points (Figure 6). In addition, when computing spectral contrast, the spectrogram is divided into seven octave-scale sub-bands (0–200, 200–400, 400–800, 800–1600, 1600–3200, 3200–8000, and >8000 Hz; Jiang et al., 2002); thus, the sub-bands are also presented (Figure 6). We found the spectral contrast values of sub-bands, whose frequency is 0–200 Hz or higher than 1600 Hz, to explain most of the variation in spectral contrast. Moreover, in the same sub-

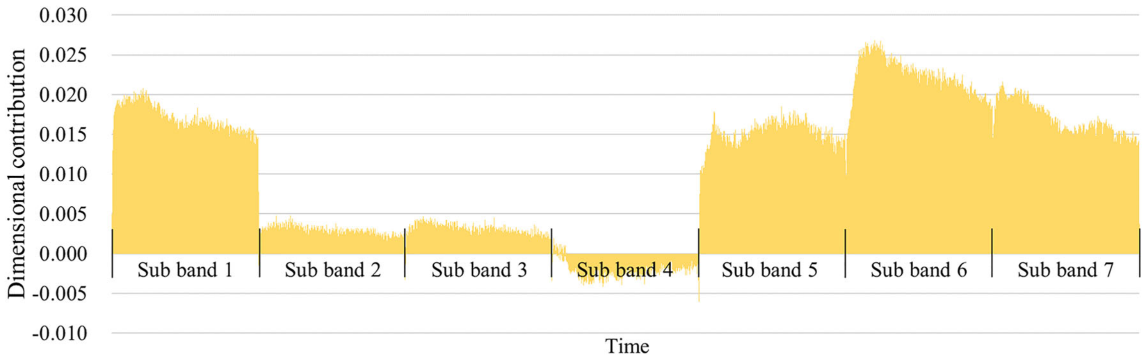

FIGURE 6 Interpretation of the first principal components analysis (PCA) component of spectral contrast

band, the earlier part of the music provides more information than the later part (e.g., sub-band 1: $t = 25.399$, $p < .001$, $d = 2.122$; sub-band 6: $t = 12.613$, $p < .001$, $d = 1.039$).

DISCUSSION

This study explored the effects of different music features on the perception of music depth in three steps. First, we analyzed the relationship between perceived depth and two high-level features (valence and arousal), and found that they were negatively correlated. Next, we investigated the linear relationship between perceived depth and nine low- or mid-level music features, including tempo, MFCC, chromagram, spectral centroid, spectral bandwidth, spectral contrast, spectral flatness, spectral roll-off, and tonnetz. Finally, we applied ML methods to explore the nonlinear links between music depth perception and music features, and the RFR-based model showed excellent prediction effect. The above processes resulted in an understanding of the relationship between depth perception and music features through a computational modeling perspective. Furthermore, the recognition models of perceived depth can provide people with music depth information automatically, which may offer some practical values.

We found that perceived depth was negatively correlated with both perceived valence and arousal, which means a deep song is more likely to be negative and low arousal. We guess that the difference in the underlying music features caused the negative correlation. For example, previous works have shown that positive music (Gabrielsson & Juslin, 2003; Juslin & Laukka, 2004) and high arousal music (Gabrielsson & Lindström, 2001) is apt to be fast, which is opposite to the dreamy music (a sub-attribute of depth; Rigg, 1964). Likewise, high arousal music is more likely to be high pitch and bright (Gabrielsson & Lindström, 2001), opposite to the deep music (see Section 2.2). Thus, our subsequent work focused on the relationship between perceived depth and music features. Of course, there may be many other reasons that can explain the above negative correlation. For instance, when entering states of absorption, individuals may pursue negative (Garrido & Schubert, 2013) and deep music, because they may both help to deactivate displeasure circuits; similar effects may lead to similar perceptions of negative and deep music. However, these assumptions still need further verification.

We then explored the direct links between perceived depth and low- and mid-level music features. Correlation analysis and the MLR-based model showed the linear relations between variables, and spectral flatness is closely related to deep music (negatively correlated with perceived depth, $r = -.553$, $p < .001$; accounted for 19.85% of the MLR-base model). Spectral flatness is a measure to quantify how much noise-like a sound is (Dubnov, 2004), and a high spectral flatness (closer to 1.0) indicates the music is similar to white noise. This means noise-like music is more likely to be perceived as music without depth. Spectral roll-off and spectral centroid also played important roles in the MLR-based model. Spectral roll-off is the 95th percentile of the power spectral distribution, and it can distinguish voiced from unvoiced speech (Kumari et al., 2010). While spectral centroid is the "balancing point" of the spectral power distribution, and it can give different results for voiced and unvoiced speech (Scheirer & Slaney, 1997). Thus, our findings support that the perception of music depth is also related to the voiced and unvoiced speech. MFCC contributed the fourth most information in the MLR-based model. MFCC are perhaps the most commonly used acoustic features in currently available speech and music recognition systems (Chen et al., 2001; Yang et al., 2018). Therefore, it is not difficult to imagine the key role of MFCC in music depth recognition.

The nonlinear relation between perceived depth and music features was also investigated by constructing computational models. The RFR-based model performed best in predicting perceived depth values; thus, nonlinear approaches may better explain the relationship between variables than linear approaches here. To interpret the RFR-based model, feature importance was then computed (Quiroz et al., 2018). The result shows that the first PCA component of spectral contrast accounted for 70.04% of the model, which is significantly different from the MLR-based model. This result means that listeners judge whether the music has depth mainly depends on the spectral contrast value of the audio, which reflects whether the music has clear, narrow-band signals (Jiang et al., 2002). This discovery may conflict with previous cognition (e.g., Nagathil et al., 2018; Rigg, 1964), so we further explored the above PCA component. Results show that some dimensions, computed in certain sub-bands (0–200 Hz or >1600 Hz), explained most of the variation of the target component. This means that the human auditory system may process the audio signal at 0–200 Hz and above 1600 Hz into specific values, which is the main basis for music depth judgement.

In addition, our work is of practical value. There is so much existing music in human life, lack of music information brings great inconvenience to music management (Kim et al., 2010; Xu et al., 2020a). This study constructed recognition models of music depth, which can automatically detect music depth information. The detected information may be used in music information retrieval (Downie, 2008; Wang et al., 2012) and music recommendation (Park et al., 2015) systems.

Notably, this study has several limitations. First, the music features considered in this study may only be a subset of the full features actually processed by listeners. More music features, such as genre information (Li & Ogihara, 2003) and lyric features (Wu et al., 2014), can be investigated in further research. Next, music stimulation in the PSIC3839 data set are all entire songs (Xu et al., 2020b), whose length may not be suitable for perceptual research. The deep content of a song may fluctuate, so we can split music into segments; the shorter the segment, the more homogeneous the depth might be. However, it may not be sufficient for the participants to evaluate music depth of too short segments. Therefore, just like what Xiao et al. (2008a) did in music emotion recognition study, finding the balance point (the appropriate segment length to evaluate music depth) is one of the future research directions. Third, this study directly used the mean value of each song in the PSIC3839 database as the independent variable of the statistical test and the ground truth of the model. However, each song may not be completely independent, because each song is randomly labeled by the same group of participants. Although most of the current music emotion databases use this method for annotation (e.g., Speck et al., 2011; Xiao et al., 2008b; Yang et al., 2007), we still need to find ways to overcome the above shortcoming in future research. Fourth, this work mainly investigated the connection between music structure and music depth, but we did not consider the role of high-level features in depth. In fact, we have found that perceived depth is negatively correlated with arousal and valence. Thus, is perceived emotion of music a mediating variable of music depth perception? Are there other high-level features that affect the music depth perception? These are important directions for future research. Fifth, Strauss (1949) described that the famous composer Richard Wagner "demanded from the conductor a proper comprehension of the basic tempo as essential to the correct interpretation of a piece of music". We believe that tempo may also be a crucial parameter with respect to the perceived depth of music. Therefore, the characteristics of temporal processing of neuro-cognitive mechanisms should also be considered in future research. Finally, considering the model interpretation, we applied three traditional ML methods for model construction. In fact, more flexible methods can be used to pursue model effects, including deep neural networks (Orjesek et al., 2019), convolutional neural networks (Mao et al., 2014), and so forth.

CONCLUSION

In this study, we explored the links between the perception of music depth and different music features. We observed that (1) perceived depth of music is negatively correlated with valence and arousal; (2) perceived depth values are related to different music features, including tempo, MFCC, chromagram, spectral centroid, spectral bandwidth, spectral contrast, spectral flatness, spectral roll-off, and tonnetz; (3) selected music features can directly predict the perceived depth of music; and (4) RFR-based music depth recognition model shows an excellent prediction result. Our work explained the inner connection between perceived depth and some music features, and also constructed models with practical value.

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CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest.

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